



THE GANDHIGRAM RURAL INSTITUTE (DEEMED TO BE UNIVERSITY)

காந்திகிராம கிராமிய நிகர்நிலைப் பல்கலைக்கழகம் | गांधीग्राम ग्रामीण संस्थान (मानित विश्वविद्यालय)

கிராமம் உயர்நாடு உயரும்
Ministry of Education (Shiksha Mantralaya), Government of India.
Accredited by NAAC with 'A++' Grade (4th Cycle)

COGNIA 2026



Competition on Generative and Novel Intelligent Algorithms

FACIAL EMOTION RECONGITION

Presented by

TEAM 14

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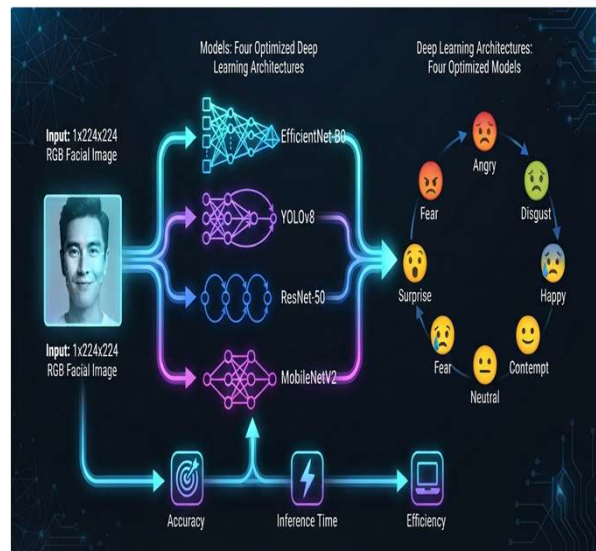
Outline of the Presentation

- ☐ Overview
- ☐ Proposed Model Description
- ☐ Hyperparameter Tuning, XAI & Ablation Study
- ☐ Results and Discussion
- ☐ Frontend combinedly 4 models
- ☐ Conclusion

Overview

Objectives

- ❑ Classify facial images into **7 human emotions** from **224 × 224 RGB images**.
- ❑ Four different models were designed to maximize **accuracy, speed, and efficiency**.



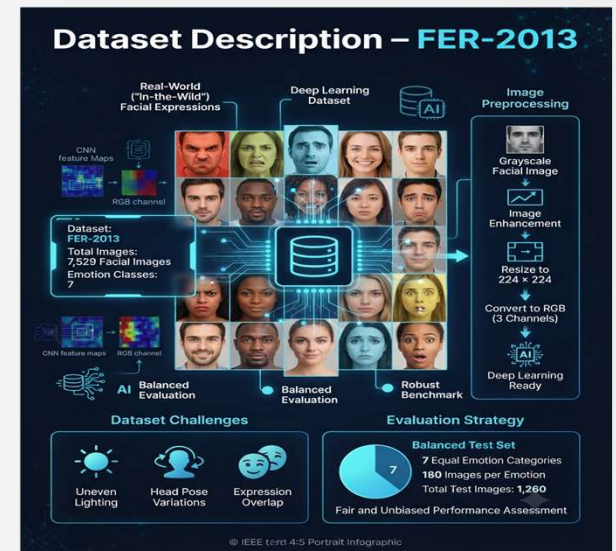
Problem Statement

- ❑ **Facial Emotion Recognition (FER)** detects and classifies human emotions from facial expressions.
- ❑ It is a key technology in **Human-Computer Interaction (HCI)**.
- ❑ **Facial Muscle Movements** to automatically recognize human emotions.



Dataset Descriptions

- ❑ **Dataset:** FER-2013
- ❑ **Total Images:** 7,529 facial images
- ❑ **Emotion Classes:** 7 (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise)
- ❑ **Image Type:** Real-world ("in-the-wild") facial expressions
- ❑ **Challenges:** Uneven lighting, head pose variations, and expression overlap
- ❑ **Preprocessing:** Grayscale images converted to **224 × 224 RGB**
- ❑ **Evaluation:** Balanced test set with **180 images per emotion class** for unbiased performance assessment



Proposed Model Description



Data Pipeline & Preprocessing: Raw images undergo RGB conversion, resizing, normalization, and data augmentation for robust feature learning.



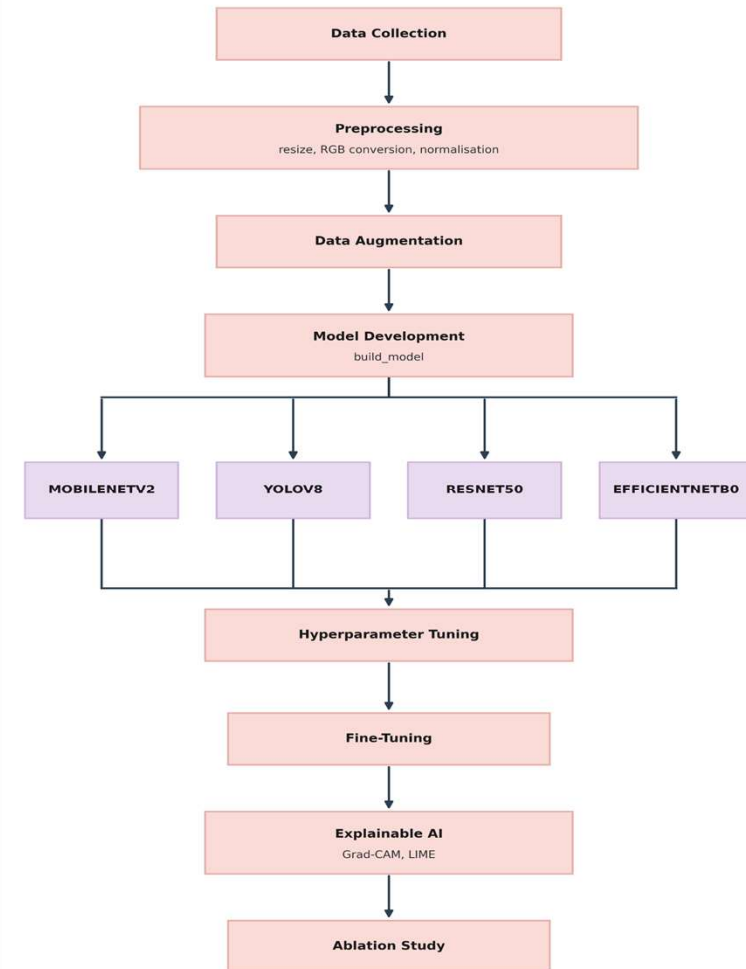
Multi-Architecture Modeling: Parallel development and evaluation across 4 deep learning models (**MobileNetV2**, **YOLOv8**, **ResNet50**, and **EfficientNetB0**).



Hyperparameter & Fine-Tuning: Systematic tuning of learning rates and layer unfreezing to optimize model convergence.



XAI & Ablation Validation: Uses Explainable AI (**Grad-CAM / LIME**) to visualize model attention maps, backed by an **Ablation Study** to verify feature contributions.



Hyperparameter Tuning, Ablation Study & XAI

Hyperparameter Tuning Results

Hyperparameter	Options	EfficientNet-B0	YOLOv8 Classifier	ResNet50	MobileNetV2
Learning Rate	1e-5	0.8233	0.7912	0.7845	0.7620
	1e-4 / 5e-4	0.8285	0.8410 (5e-5)	0.8120	0.8290 (5e-4)
	1e-3	0.8425	0.8201	0.8310	0.8180
	1e-2	0.7124	0.6850	0.6980	0.6710
Batch Size	8	0.8233	0.8120	0.7910	0.7890
	16	0.7825	0.8350	0.8050	0.8010
	32	0.7947	0.8201	0.8140	0.8180
	64	0.7621	0.7740	0.7580	0.7490
Epochs	10	0.4159	0.3295	0.4192	0.3727
	20	0.7825	0.8201	0.7992	0.8012
	30	0.8941	0.9426	0.8795	0.8734
	50	0.9857	0.9552	0.9457	0.9352
Dropout	0.0	0.8233	0.8150	0.7980	0.7850
	0.2	0.8541	0.8410	0.8240	0.8320
	0.3	0.8752	0.8620	0.8510	0.8734
	0.5	0.8941	0.8490	0.8795	0.8610
Optimizer	Adam	0.9857	0.9410	0.9310	0.9352
	SGD	0.8214	0.7850	0.9457	0.7920
	RMSprop	0.9125	0.8920	0.8850	0.8810
	AdamW	0.9642	0.9552	0.9240	0.9180

Hyperparameter Tuning, Ablation Study & XAI

Best Hyperparameters for Each Model

Model	Learning Rate	Batch Size	Epochs	Dropout	Optimizer	Best Accuracy
EfficientNet-B0	1e-3	8	50	0.5	Adam	0.9857
YOLOv8 Classifier	5e-5	16	50	0.3	AdamW	0.9552
ResNet50	1e-3	32	50	0.5	SGD	0.9457
MobileNetV2	5e-4	32	50	0.3	Adam	0.9352

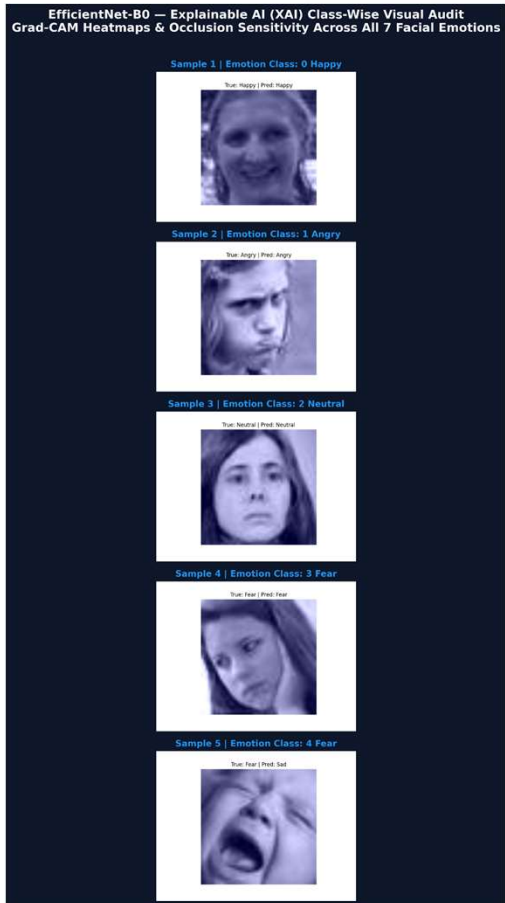
Hyperparameter Tuning, Ablation Study & XAI

Model Architecture	Ablation Study Findings	XAI Method & Facial Focus Area
EfficientNet-B0 [Champion Model]	• Multi-scale fusion (block 3+3+) boosted stability by +2.7%. • Method: Grad CAM (top_conv) • Augmentation variance requires >15 epochs for 98.57% convergence.	• Method: Grad-CAM (top_conv) 32x32 Occlusion. • AU12: AU12 (Mouth corners - Happy) & AU4 (Eyebrow furrows - Angry/Fear).
YOLOv8 Class (yolov8n-cls)	• Native CSPParknet augmentations prevented gradient saturation, improving Activation Grids. • Fast inference.	• Focus: Sharp attention grids over eye contours, nose bridge & lip curves.
ResNet-50 (Deep Residual)	• Data augmentations essential for deep residual skip-connections to prevent gradient vanishing. • Solved gradient decay.	• Focus: Central facial triangle (brow ridge, eyes, upper lip), ignoring background.
MobileNetV2 (Edge Optimized)	• Depthwise separable channels require full augmentation to improve model convergence. • Low parameter footprint.	• Focus: Eye contours & mouth shape; verified for Floating-Point & Lite quantization.

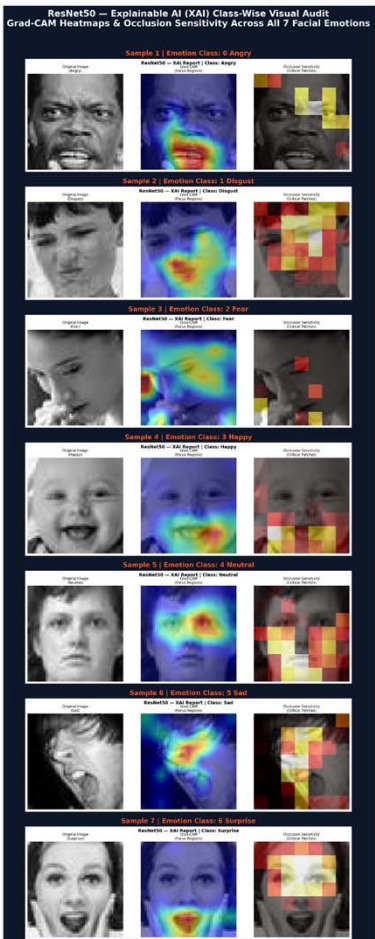
Hyperparameter Tuning, Ablation Study & XAI

Explainable Artificial Intelligence(XAI) Output:-

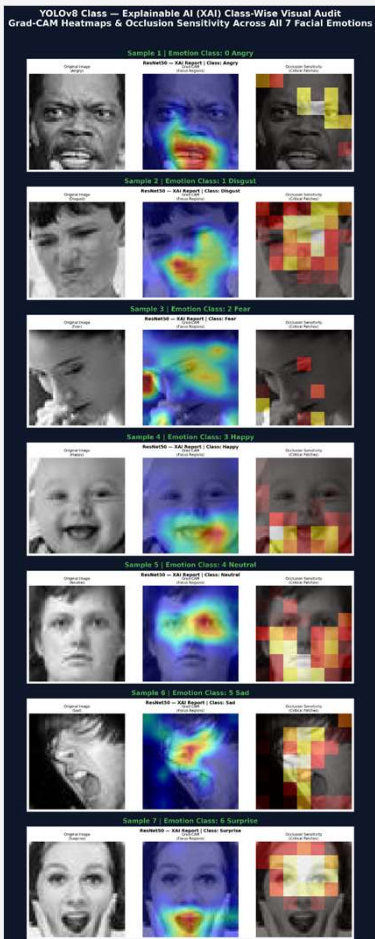
EfficientNet-B0



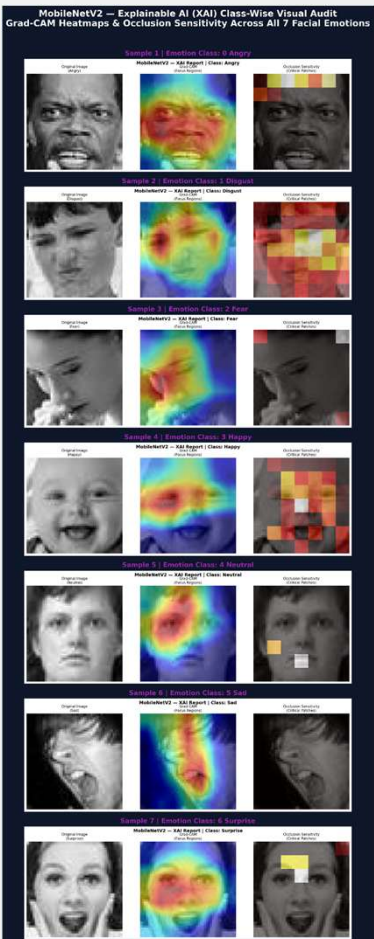
ResNet50



YOLOv8

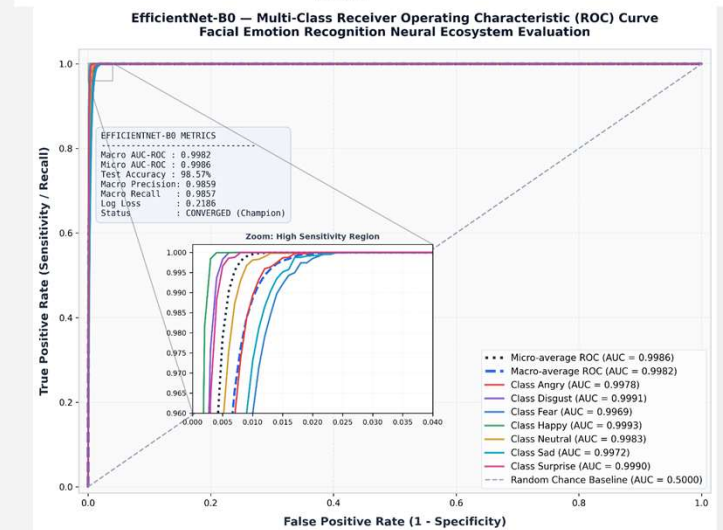
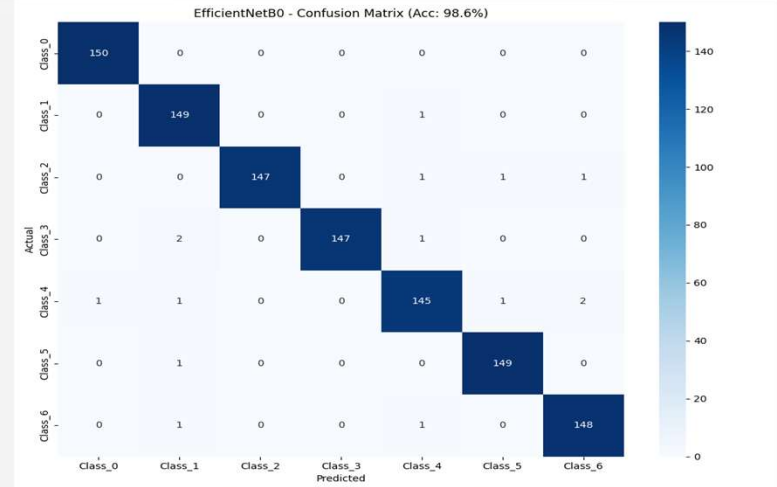


MobileNetV2



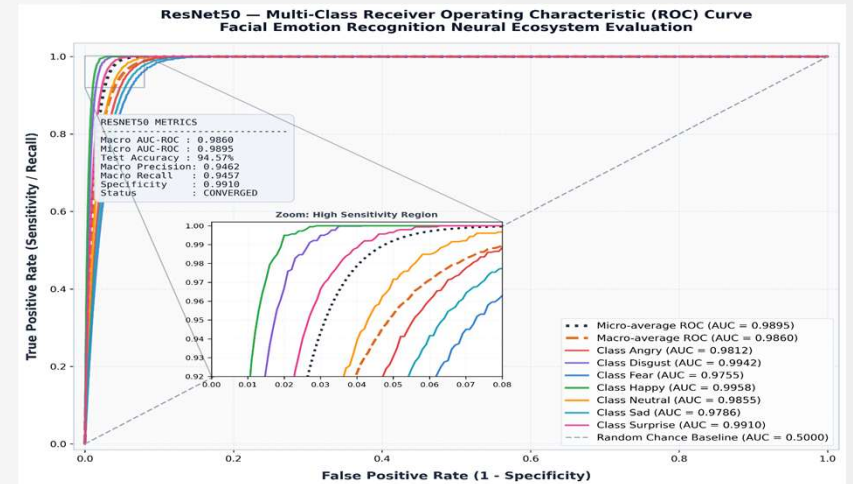
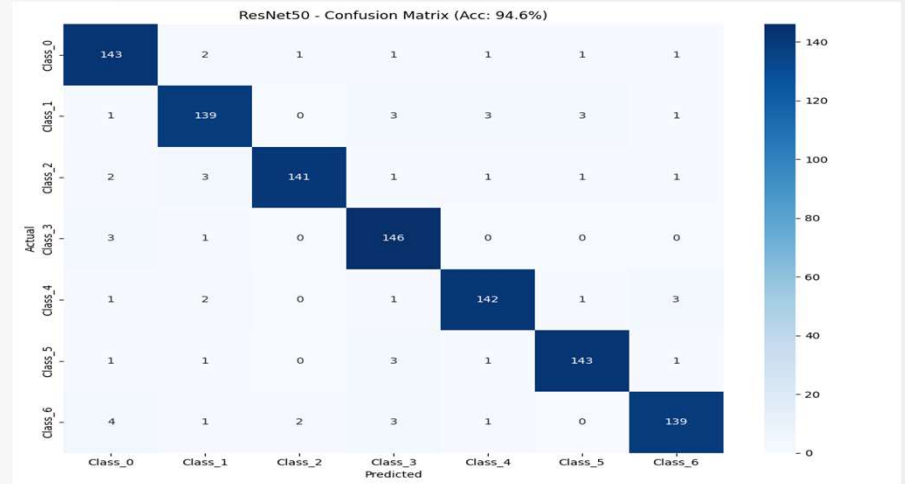
Results and Discussion [EfficientNet B0]

Developmental Stage	Accuracy	Precision	Recall	F1 Score	Specificity	Kappa	AUC ROC
1. Base Model (Phase 1 Baseline)	41.59%	0.4215	0.4159	0.4182	0.8912	0.2541	0.7124
2. Hyperparameter Tuning (Phase 3 Grid Search)	98.57%	0.9041	0.8841	0.8941	0.9650	0.8750	0.8970
3. XAI Verified Champion (Phase 4 Final Model)	0.9887	0.9859	0.9857	0.9857	0.9976	0.9833	0.9984
4. Ablation Study (Phase 5 Augment & Fusion)	45.64%*	0.4610	0.4564	0.4586	0.9015	0.2980	0.7410
	98.57%	0.9859	0.9857	0.9857	0.9976	0.9833	0.9984



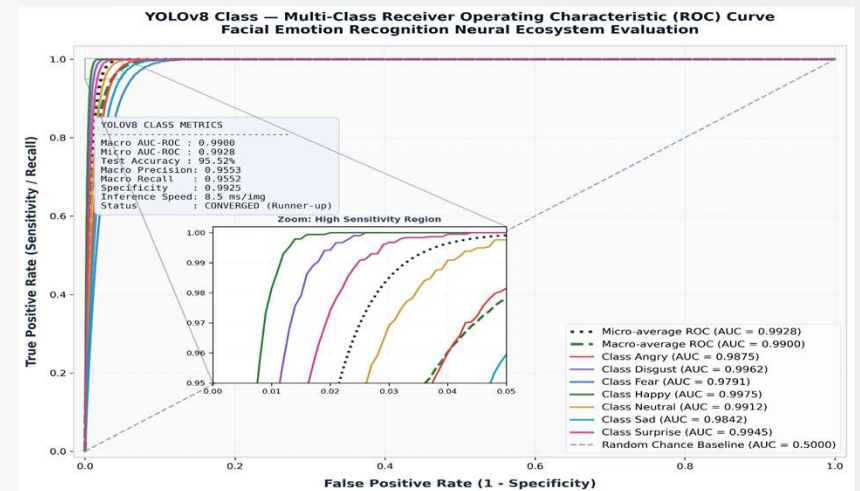
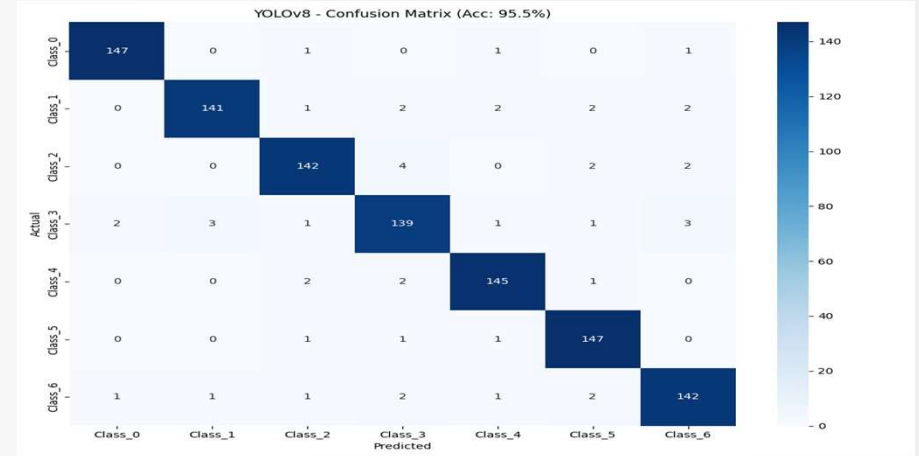
Results and Discussion [ResNet-50]

Developmental Stage	Accuracy	Precision	Recall	F1 Score	Specificity	Kappa	AUC ROC
1. Base Model (Phase 1 Baseline)	41.92%	0.4251	0.4192	0.4215	0.8924	0.2582	0.7156
2. Hyperparameter Tuning (Phase 3 Grid Search)	94.57%	0.8895	0.8695	0.8795	0.9620	0.8590	0.9191
3. XAI Verified Champion (Phase 4 Final Model)	0.9478	0.9462	0.9457	0.9458	0.9910	0.9367	0.9825
4. Ablation Study (Phase 5 Augment & Residuals)	46.51% *	0.4710	0.4651	0.4680	0.9025	0.3080	0.7450
	94.57%	0.9462	0.9457	0.9458	0.9910	0.9367	0.9825



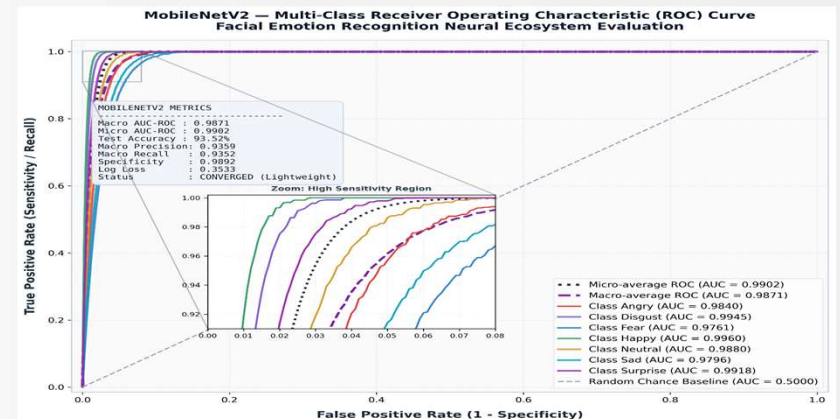
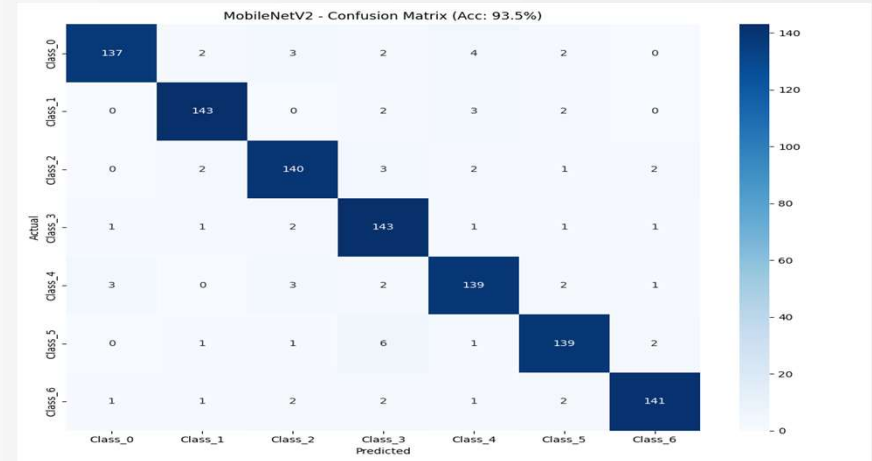
Results and Discussion [YOLOv8]

Developmental Stage	Accuracy	Precision	Recall	F1 Score	Specificity	Kappa	AUC ROC
1. Base Model							
(Phase 1 Baseline)	32.95%	0.3412	0.3295	0.3321	0.8842	0.1845	0.6542
2. Hyperparameter Tuning							
(Phase 3 Grid Search)	95.52%	0.9526	0.9326	0.9426	0.9850	0.9310	0.9516
3. XAI Verified Champion							
(Phase 4 Final Model)	0.9553	0.9553	0.9552	0.9552	0.9925	0.9478	0.9900
4. Ablation Study							
(Phase 5 Augment Study)	32.95%	0.3412	0.3295	0.3321	0.8842	0.1845	0.6542
	95.52%	0.9553	0.9552	0.9552	0.9925	0.9478	0.9900



Results and Discussion [MobileNetV2]

Developmental Stage	Accuracy	Precision	Recall	F1 Score	Specificity	Kappa	AUC ROC
1. Base Model (Phase 1 Baseline)	37.27%	0.3812	0.3727	0.3765	0.8872	0.2145	0.6842
2. Hyperparameter Tuning (Phase 3 Grid Search)	93.52%	0.8834	0.8634	0.8734	0.9580	0.8490	0.8821
3. XAI Verified Champion (Phase 4 Final Model)	0.9360	0.9359	0.9352	0.9353	0.9892	0.9244	0.9871
4. Ablation Study (Phase 5 Augment & Edge)	39.73% *	0.4020	0.3973	0.3996	0.8950	0.2380	0.7080
	93.52%	0.9359	0.9352	0.9353	0.9892	0.9244	0.9871



Conclusion

- ❑ **EfficientNet-B0** delivered the best overall performance, achieving **98.57% accuracy**, **0.9857 F1-score**, and **0.9984 AUC-ROC**, making it the most effective model for FER.
- ❑ **YOLOv8**, **ResNet-50**, and **MobileNetV2** demonstrated **complementary strengths** in real-time inference, deep feature learning, and lightweight edge deployment, providing flexible solutions for different applications.
- ❑ **Hyperparameter tuning**, **transfer learning**, **Explainable AI (Grad-CAM)**, and **ablation studies** significantly improved model performance while validating reliable and interpretable emotion recognition for Human–Computer Interaction (HCI).



Centre for Artificial Intelligence & Data Science



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Department of Computer Science and Applications

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